

UAVXGS — Ground Station User Guide

Overview

UAVXGS presents five main windows plus several instrument panels. This guide describes each window's function and how to engage with it.

1. Main Window — Instrument Panel

The primary window that opens on launch. It has four areas:

Toolbar (top)

Control	Function
Connect	Opens serial port selector; click again to disconnect
COM:	Serial port selector — auto-detect or manual /dev/ttyXXX / COMx
Baud:	Baud rate (115200 for UAVXArmQ)
Params	Opens the Parameter Window
Nav	Opens the Navigation / Mission Planning Window
Calib	Opens the Calibration Window
Misc	Opens the Raw Packet Monitor
Flash	Opens the DFU Firmware Flasher
KML	Checkbox — enables KML track logging
Log	Checkbox — enables CSV telemetry logging
BB	Checkbox — starts black-box dump from FC

Attitude Sphere (left, large)

3D artificial horizon showing roll and pitch. The aircraft icon is fixed; the horizon tilts. A compass rose rotates around the rim showing magnetic heading.

Below the sphere: - **Exec bar** — CPU load bargraph - **Revision label** — firmware name (e.g. "UAVX r16" or airframe name)

Altitude Display (below sphere)

Large numeric altitude readout in metres (Kalman-filtered baro+acc). Beside it: - **ROC** — rate of climb in m/s - Sub-labels show the source (Altitude / ROC)

Instrument Panels (right)

State & Battery Row - Flight state (Idle / InFlight / Monitor) - Nav state (PIC / Hold / RTH / WP) - Current waypoint number - Alarm status - Battery voltage, current draw, consumed mAh

Controls Row - Four miniature bargraphs: Throttle, Roll, Pitch, Yaw - Green bars show normalised output (0-100%)

IMU Box - Angles: Roll, Pitch, Yaw in degrees - Gyros: Raw rates in rad/s - Accels: L/R, F/B, D/U in m/s² - MPU temperature - Acc confidence (0-100%)

Navigation Box - Next waypoint ID, bearing, distance, cross-track error - Guidance bearing, elevation, home distance, hint - Wind speed and direction (when WindEstValid) - Magnetic variation (WMM)

Altitude Box - Kalman-filtered altitude, ROC - Baro variance, accel variance, bias variance - Desired altitude (target)

GPS Box - Latitude, Longitude (decimal degrees) - Ground speed (m/s), course over ground (°) - Satellites, HDOP, fix type - 3D speed - Altitude (GPS MSL)

Flags Box - LED-style indicators: Armed, InFlight, AngleMode, AltHold, AltActive, GPS, Home-Set, WindEstValid, MagGood, HaveMagCal, UsingGPSAlt, HaveRangeFinder

Voice Feedback Level - Dropdown in the status bar: Off, Errors, Warnings, Info, All - Uses espeak (Linux) / NSSpeechSynthesizer (macOS) / SAPI5 (Windows)

2. Parameter Window

Opened via the **Params** toolbar button.

Layout

Tabular view with parameter rows grouped by category tabs:

Tab	Parameters
Basic	Airframe type, Rx mode, battery, emulation
Gains	Roll/Pitch/Yaw PIDs, feedforward, rate limits
Rates	Max roll/pitch/yaw rates, throttle curve
Altitude	Altitude hold gains, baro config, KF tuning
Navigation	Waypoint radii, velocity limits, bank angle limits
Misc	Serial ports, telemetry rate, speaker, LED config
Tuning	Cruise throttle, accel/decel gains, vibration damping

Operations

Action	How
Read	Reads all 128 params from FC
Write	Writes all changed params to FC
Save	Saves current set to .af file
Load	Loads a .af file into the table (not written to FC until Write)
Commit	Writes + conditions + flashes + resets FC
Edit	Double-click a value cell, type new display value, press Enter
Revision	Displays current firmware revision at top of window

Display vs FC Units

Parameters are shown in display units (degrees, percent, metres, etc.) and converted to FC-native units (radians, 0-1, etc.) on write. The conversion multiplier is shown in the table header.

3. Navigation Window

Opened via the **Nav** toolbar button.

Map View

- OpenStreetMap tiles loaded via `QWebEngineView`
- Aircraft position shown as a blue aircraft icon with heading arrow
- Home position shown as a green H
- Waypoints shown with ID labels and connecting lines

Mission Controls

Action	How
Read	Downloads the current mission from FC
Write	Uploads the current mission to FC
Clear	Removes all waypoints
Load	Opens a mission file
Save	Saves current mission + map screenshot
Add WP	Adds a waypoint at the map centre
Delete WP	Removes selected waypoint
GoTo	Single-click on map creates a 1-point GoTo mission
Right-click menu	Set waypoint type: VIA, ORBIT, PERCH, SETPOI, SURVEY

Waypoint Types

Type	Behaviour
VIA	Fly through at altitude, pause for T seconds
ORBIT	Orbit at radius R with velocity V
PERCH	Land at waypoint, then take off and resume
SETPOI	Sets a point of interest — aircraft points camera here
SURVEY	Emits periodic pulse on Aux1 pin

Emulation

When no FC is connected, the GCS auto-starts emulation mode. The FC simulator generates synthetic GPS and flight data. The Nav window shows a default mission that can be edited and tested without hardware.

4. Calibration Window

Opened via the **Calib** toolbar button.

Sensors

Button	Calibration	What to Do
Calibrate IMU	Acc + Gyro	Place aircraft level, click, wait for green
Calibrate Acc	Accelerometer only	Same as IMU
Calibrate Gyro	Gyroscope only	Aircraft still, click, wait
Calibrate Mag	Magnetometer	Rotate aircraft in all orientations, click, wait
Calibrate Level	Level trim	Aircraft level, click

Indicators

- Each sensor shows a status LED: Red (uncalibrated) → Yellow (in progress) → Green (done)
- Live sensor readback during calibration
- All calibration buttons are disabled when aircraft is armed

Utility Buttons

Button	Function
Force Defaults	Resets all parameters to factory defaults
Init GPS	Re-initialises the GPS module
GPS Passthrough	Enters GPS configuration passthrough mode
Bootloader	Jumps to DFU bootloader (for firmware flash)

5. Misc Window — Raw Packet Monitor

Opened via the **Misc** toolbar button.

Display

A scrolling table showing every received packet:

Column	Content
Tag	Decimal tag number
Name	Human-readable name (Flight, Tuning, Param, etc.)
Dir	R (Received) or T (Transmitted)
Body	Hex dump of raw packet body

Use Cases

- Protocol debugging — verify all expected tags are arriving at correct rate
 - Bandwidth analysis — count packets per second by tag
 - New feature development — check that new packet parsers work correctly
 - Signal integrity — look for checksum errors or partial packets
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6. DFU Flasher Window

Opened via the **Flash** toolbar button.

Function

Uploads firmware to the STM32 flight controller via the UART DFU bootloader. No external programmer (ST-Link) required.

Workflow

1. Select the firmware .bin file via the file picker
 2. Put the FC in DFU mode (hold boot button, power cycle, or click “Enter DFU” in the flasher)
 3. Click **Flash**
 4. Progress bar shows upload progress
 5. Click **Reset** to restart the FC with new firmware
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7. Status Bar (bottom of main window)

Element	Function
Status message	Brief text showing current operation / errors
Debug	Checkbox to toggle debug messages
Level	Debug verbosity: Info, Warnings, Errors, All
Speech	Voice feedback level selector
Connected indicator	Red/green dot with connect status